You said:

Vecotr DB vs Other Database explain why we used vector db for RAG

ChatGPT said:

Vector Database vs Other Databases — Why Do We Use Vector DB in RAG?

1. What is a Vector Database?

Definition: A Vector Database is a specialized database designed to store, index, and search vectors (numerical representations of data called embeddings) based on their similarity in meaning.

In Generative AI, text, images, and other data can be converted into numerical vectors using an embedding model. A vector database stores these vectors along with the original content and metadata, allowing us to retrieve information that is semantically similar to a user's query.

Example:

Suppose we store this sentence in a vector database:

> "Python is a programming language used for AI and Machine Learning."

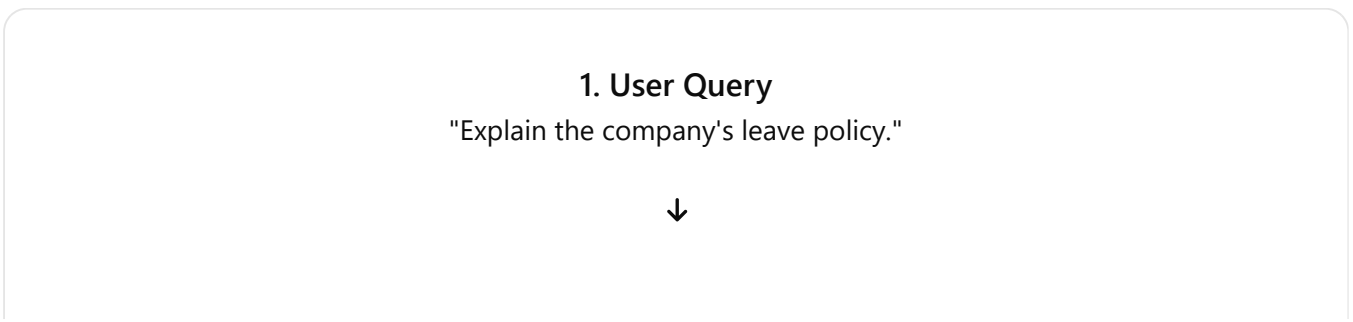
A user asks:

> "Which language is useful for building AI applications?"

Even though the words are different, the vector database can identify that the question is semantically similar to the stored sentence.

2. What is RAG, and where does Vector DB fit?

RAG stands for **Retrieval-Augmented Generation**. It is a technique that retrieves relevant information from external data and provides it to an LLM to generate a more grounded answer.



2. Embedding Model
Converts the query into a numerical vector.

↓

3. Vector Database
Finds semantically similar document chunks.

↓

4. Relevant Context + User Query
Retrieved information is passed to the LLM.

↓

5. LLM Generates an Answer
Answers using the retrieved company policy.

Important: The vector database retrieves relevant information; the LLM generates the final answer. The vector database does not generate the response itself.

3. Vector DB vs Other Databases

Different databases are designed for different types of data and queries. The biggest difference is **how they search for information**. Microsoft Le... +1

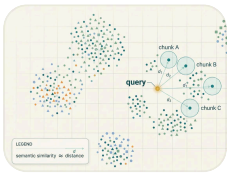
Feature	Vector DB	SQL / Relational DB	NoSQL DB	Traditional Search Engine
Examples	Chroma, Pinecone, Qdrant	MySQL, PostgreSQL	MongoDB, Cassandra	Elasticsearch, OpenSearch
Data storage	Vectors + metadata	Rows and columns	Documents, key-value, etc.	Searchable documents and indexes
Main search	Semantic similarity	Exact values, ranges, joins	Depends on database and query	Primarily text search; some support vectors
Understands meaning	Yes, through embeddings	Not by default	Not by default	Keyword search by default; can support semantic search
Best for	RAG, semantic search	Banking, users, orders	Flexible application data	Full-text search and log analysis

RAG suitability	Excellent for vector retrieval	Possible with vector extensions	Possible if vector search is supported	Excellent when semantic or hybrid search is supported

Note: These database categories can overlap. PostgreSQL supports vector search through extensions such as `pgvector`, and Elasticsearch/OpenSearch can support both keyword and vector search. About Vector... +1

4. Why do we use a Vector DB in RAG?

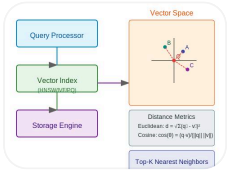
There are six main reasons.



1. Semantic Search

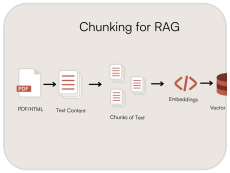
It finds information based on meaning rather than requiring the exact same words.

Example: "How can I get a refund?" can retrieve a document titled "Product return and reimbursement policy."



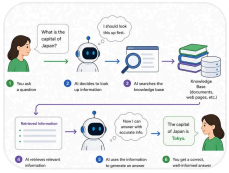
2. Efficient Similarity Search

Specialized vector indexes can quickly identify nearby vectors across large collections without comparing every vector in every search. The exact performance depends on the database, index, and dataset size.



3. Handles Document Embeddings

Documents are split into chunks, converted into embeddings, and stored with their text or references. This allows RAG to retrieve relevant sections of PDFs, reports, manuals, and other documents.



4. Provides External Knowledge to LLMs

It retrieves relevant context from your own documents, helping an LLM answer questions about private or specialized information it may not know.



5. Metadata Filtering

Many vector databases let you filter results by metadata, such as document category, department, date, or source, before passing results to the LLM.



6. Scalability

Vector databases can support large collections of embeddings and high volumes of similarity queries, depending on the chosen product and configuration.

IBM +1

5. Real-time example: College chatbot using RAG

Suppose you build a chatbot that answers questions about your college's PDFs.

Your documents contain this information:

> MCA students must complete the project submission before the announced deadline.

A student asks:

> "When should I submit my MCA project?"

Here is what happens:

Step	Process
1	The system reads the college PDF.
2	It splits the PDF into smaller text chunks.
3	An embedding model converts each chunk into a vector.
4	The vector database stores the vectors with the corresponding text or document references.
5	The student's question is converted into a query vector.
6	Similarity search retrieves the relevant project-submission information.
7	The LLM uses that information to formulate an answer.

The chatbot can answer using the college document instead of relying only on the LLM's general knowledge.

6. Why not use only a MySQL or MongoDB database?

You *can* use traditional databases in a RAG system, but their suitability depends on their capabilities.

- **MySQL:** Excellent for structured information such as student IDs, marks, and fees. Basic SQL queries do not inherently provide semantic similarity search.

- **MongoDB:** Excellent for flexible document storage. Vector search is possible with supported vector-search functionality.
- **PostgreSQL:** Can store normal relational data and embeddings using `pgvector`, so you may not need a separate vector database.
- **Chroma or Qdrant:** Purpose-built options for storing embeddings and performing vector similarity search.

For a small project, a dedicated vector database is not always necessary. You can use an existing database with vector-search support, or even perform a simple in-memory similarity search for a small dataset. About Vector... +1

7. Popular Vector Databases used in RAG



Chroma

Good for learning RAG and building small to medium-sized prototypes.



Pinecone

Managed vector database service for production semantic-search and RAG applications.



Qdrant

Vector search with metadata filtering and options for deployment and scaling.



FAISS

A vector similarity-search library, rather than a full standalone database. Useful for efficient similarity search and experimentation.

8. Advantages and disadvantages

Advantages	Disadvantages
Finds semantically relevant information	Requires an embedding model for embedding-based search
Efficient vector similarity search	Embedding storage and indexing consume resources
Useful for RAG and recommendation systems	Retrieval quality depends on chunking, embeddings, and indexing
Can scale to large vector collections	Dedicated infrastructure may increase cost and complexity
Often supports metadata filters	May not replace relational databases for transactions and joins

9. Interview answer — remember this

Question: Why do we use a Vector Database in RAG instead of a traditional database?

Answer: We use a vector database in RAG because it stores numerical embeddings and retrieves relevant information based on semantic similarity. Traditional databases are primarily optimized for structured queries, exact matches, filtering, and transactions. A vector database helps retrieve relevant document chunks even when the user's question uses different words from the original document. The retrieved context is then passed to the LLM to generate a more relevant, context-aware answer.

One-line trick to remember:

- SQL DB = Find exact records.
- Vector DB = Find similar meaning.
- RAG = Retrieve relevant information + generate an answer.